Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India
(Autonomous College Affiliated to University of Mumbai)

ESE Examination
March 2024

Max. Marks: 100 Duration: 3hr.
Class: T.E. (VI) CSE AIML/DS Course Code: AI306/DS306
Name of the Course: Distributed Computing

Instructions:
(1) All questions are compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Q. No.		Max Marks	CO-BL
Q1 A	Apply your knowledge of distributed information systems to construct a scenario where a multinational corporation seeks to develop a distributed Transaction Processing System (TPS) for managing its global operations. Choose appropriate architectural components and technologies to build this system, ensuring scalability, fault tolerance, and efficient transaction processing across distributed nodes. Provide a detailed explanation of your chosen architectural decisions and justify their suitability for the corporation's requirements.	10	1-3
Q1 B	Examine the efficiency and advantages of the workstation-server model within rendering farms employed by animation studios. Discuss two inherent limitations of the workstation-server model and propose potential mitigation strategies. Ensure your response is comprehensive and well-structured.	10	1-4
Q2 A	Consider a banking application where multiple tellers serve customers in a thread-based system. Describe how locks (mutexes) and semaphores can be used to ensure synchronization and efficient resource management in this scenario. Include a discussion on how locks enforce exclusive access to tellers by customers and how semaphores regulate the number of customers allowed to access tellers simultaneously. Additionally, discuss the benefits of employing locks and semaphores in maintaining transaction integrity and preventing overcrowding at teller windows.	10	3-3
Q2 B	Examine how code migration works in distributed computing systems and how weak mobility helps the gaming platform send new game modules to players' devices for gameplay. Discuss the pros and cons of weak mobility for the gaming platform, considering things like how easy it is to use, how efficiently it uses resources, and whether it makes the gaming experience better.	10	2-

Q3 A	Discuss Cristian's algorithm within distributed computing to assess its significance in attaining time synchronization among distributed nodes with the help of diagram. List the advantages and limitations of Cristian's algorithm in achieving precise time synchronization in distributed systems. Calculate the round-trip time (RTT), delay, and synchronization time for the provided data. T0 (node 1): 10 hr:25 min:10 sec T1 (node 2): 10 hr:25 min:14 sec Ts (server): 10 hr:25 min:13 sec	10	3-3
Q3 B	Compare and contrast the following Physical Clock Synchronization algorithms with illustrative examples: 1. Berkeley Algorithm 2. NTP	10	3-4
Q4 A	Discuss Clock Synchronization in wireless networks. Explain critical path determination in RBS with suitable diagrams.	10	3-3
Q4 B	Describe the following Election algorithms in detail with diagrams: 1. Bully Algorithm 2. Ring Algorithm **OR** Highlight the significance of Load Balancing in distributed computing with a real-world example. Specify the Goals, Types and Strategies of load balancing.	10	3-2
Q5 A	Evaluate Two-Phase Commit and Three-Phase Commit with neat diagrams. Highlight the advantages of Three-Phase Commit.	10	2-3
Q5 B	Considering fault tolerant distributed system design, comment on the following: 1. Failure Masking by redundancy 2. Flat groups and Hierarchical groups in context of process resilience.	10	4-3

Synoptic

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Q. No.		CO-BL
Q1 A	Application of distributed information systems concepts: 3 marks Construction of a scenario: 2 marks Selection of architectural components and technologies: 2 marks Development of design considerations: 2marks Justification of architectural decisions: 1mark	1-3
Q1 B	Examination of efficiency and advantages of the workstation-server model: 4 marks Discussion of two inherent limitations of the workstation-server model: 4 marks Proposal of potential mitigation strategies: 2 marks	1-4
Q2 A	Explanation of locks (mutexes) in the banking application scenario: 4 marks Explanation of semaphores in the banking application scenario: 4 marks Discussion on benefits of using locks and semaphores for synchronization: 2 marks	3-3
Q2 B	Explanation of code migration and weak mobility: 3 marks Explain how weak mobility helps the gaming platform transfer new game modules: 3 marks. Evaluation of benefits and limitations of weak mobility for the gaming platform: 4 marks	2-3
Q3 A	Diagram illustrating Cristian's algorithm operation: 5 marks. Advantages and limitations discussion: 2 marks Problem solving (RTT, Delay, Synchronization time): 3 marks.	3-3
Q3 B	Compare and contrast the following Physical Clock Synchronization algorithms with illustrative examples: 1. Berkeley Algorithm 2. NTP Berkeley Algorithm: 3 marks NTP: 3 marks Examples: 4 marks (2 marks each)	3-4
Q4 A	Discuss Clock Synchronization in wireless networks. Explain critical path determination in RBS with suitable diagrams. Reference Broadcast synchronization (RBS): 5 marks Critical path in regular **VS** RBS diagram: 5 marks	3-3

Q4 B	Describe the following Election algorithms in detail with diagrams: 1. Bully Algorithm 2. Ring Algorithm Bully Algorithm: 4 marks. Ring Algorithm: 4 marks. Diagrams: 2 marks **OR** Highlight the significance of Load Balancing in distributed computing with a real-world example. Specify the Goals, Types and Strategies of load balancing. Significance of load balancing: 6 marks Goals, Types and Strategies of load balancing: 4 marks	3-2
Q5 A	Evaluate Two-Phase Commit and Three-Phase Commit with neat diagrams. Highlight the advantages of Three-Phase Commit. Two-Phase Commit: 3 Three-Phase Commit: 3 Diagrams: 3 Advantages of Three-Phase Commit: 1	2-3
Q5 B	Considering fault tolerant distributed system design, comment on the following: 1. Failure Masking by redundancy 2. Flat groups and Hierarchical groups in context of process resilience. Failure Masking by redundancy: 5 marks Flat groups and Hierarchical groups: 5 marks	4-3